

Clock Recovery

Joe Leacy

May 25, 2026

1 Modified Godard Timing Recovery

The transmitter and receiver clocks are not aligned: the ADC samples at times $t_n = nT_s + \tau$, but the symbols sit at nT_s (where $T_s = T/\eta$ is the input sample period and η is the input oversampling, $\eta = 2$ here). The timing offset τ drifts slowly with the sampling-frequency offset (SFO) and must be removed before symbol decisions can be made. The Modified Godard algorithm of Jostén et al. both estimates and corrects τ entirely in the frequency domain, so it sits naturally next to the CD equalizer where the signal is already in FFT bins.

1.1 Why the Estimator Works: the Clock Tone

A pulse-shaped signal with root-raised-cosine spectrum and roll-off β extends from $-\frac{1+\beta}{2T}$ to $+\frac{1+\beta}{2T}$. Its spectrum has support over a band of width $\frac{1+\beta}{T}$, which is wider than $\frac{1}{T}$ by exactly $\frac{\beta}{T}$ — the *excess bandwidth*. This means that if you take the spectrum $Y(f)$ and overlay the shifted copy $Y(f - \frac{1}{T})$, the two pieces overlap in two narrow bands of width $\frac{\beta}{T}$ sitting on the upper and lower spectral shoulders. Inside these overlap bands, both factors are non-zero; outside, at least one is zero.

Now introduce a timing offset τ . A delay in time multiplies the spectrum by a linear phase,

$$Y_\tau(f) = Y(f) e^{-j2\pi f\tau}. \quad (1)$$

Form the product of the upper-shoulder bin with the conjugate of the lower-shoulder bin — equivalently, multiply $R(k)$ (the FFT bin at frequency f_k) by $R^*(k + N/2)$ (the bin shifted by $1/T$, which is $N/2$ bins for $\eta = 2$):

$$R(k) R^*\left(k + \frac{N}{2}\right) = Y(f_k) Y^*\left(f_k - \frac{1}{T}\right) \cdot e^{-j2\pi f_k \tau} \cdot e^{+j2\pi(f_k - 1/T)\tau}. \quad (2)$$

The two timing-offset phases combine to leave a single factor that depends only on τ :

$$R(k) R^*\left(k + \frac{N}{2}\right) = \underbrace{Y(f_k) Y^*\left(f_k - \frac{1}{T}\right)}_{\text{real} \geq 0 \text{ in the overlap}} \cdot e^{-j2\pi\tau/T}. \quad (3)$$

This is the key fact. The pulse-shape factor is real and positive on the spectral shoulders (because Y and the shifted Y^* are matched RRC skirts that combine to a non-negative real number), so it just acts as a positive weight. Every term in the overlap band has the

same complex phase $-2\pi\tau/T$ — the so-called *clock tone*. Summing across the overlap band coherently combines the signal energy without altering this phase:

$$S = \sum_{k \in \mathcal{B}} R(k) R^*(k + \frac{N}{2}) \approx A e^{-j2\pi\tau/T}, \quad \mathcal{B} = \left\{ k : \frac{(1-\beta)N}{2\eta} \leq k < \frac{(1+\beta)N}{2\eta} \right\}, \quad (4)$$

where $A > 0$ is the in-band signal energy. Bins outside $\mathcal{B}$ are excluded because one factor is essentially zero there, so they would add only noise without contributing signal phase. The timing offset falls straight out of the argument:

$$\boxed{\hat{\tau}/T = -\frac{1}{2\pi} \arg(S)} \quad (5)$$

(The sign is convention-dependent — in the code the sign is absorbed into the phase-ramp formula.) Because $\arg$ lives on $(-\pi, \pi]$, each per-block estimate is in $(-T/2, T/2]$; a running unwrap across blocks recovers SFO drift larger than $T/2$ over the record.

Why $\arg$ rather than Im . The original Godard derivation uses $\text{Im}\{S\} = A \sin(2\pi\tau/T)$ as a feedback error signal. That works in a closed loop because the loop drives τ toward zero where $\sin(2\pi\tau/T) \approx 2\pi\tau/T$ and the small-angle approximation holds. In a feedforward architecture there is no loop — the value you compute is what gets applied directly. $\text{Im}\{S\}$ is non-linear in τ (it folds back past $\pm T/4$ and saturates), and its magnitude scales with the signal power A . $\arg(S)/(2\pi)$ is linear in τ over the entire $(-T/2, T/2]$ period and discards A , so it can be used directly without a loop filter.

1.2 Applying the Correction: Frequency-Domain Phase Ramp

Once $\hat{\tau}$ is known, the correction reuses the same block FFT. From the Fourier shift theorem,

$$\mathcal{F}\{x(t - \hat{\tau})\}(f) = X(f) e^{-j2\pi f \hat{\tau}}, \quad (6)$$

so resampling at $t = nT_s - \hat{\tau}$ corresponds to multiplying every FFT bin by a linear phase ramp. Discretising for an N -point DFT with sample period T_s , the frequencies are $f_k = k/(NT_s)$ and the delay in input samples is $\hat{\tau}_s = \hat{\tau}/T_s = \eta(\hat{\tau}/T)$. The correction is therefore one complex multiply per bin,

$$\boxed{R_{\text{corr}}(k) = R(k) \exp\left(-j \frac{2\pi k_{\text{idx}} \hat{\tau}_s}{N}\right)} \quad (7)$$

followed by an IFFT to return to the time domain.

Signed bin index. k_{idx} in eq. (7) is the *signed* FFT bin index

$$k_{\text{idx}} = \begin{cases} k & 0 \leq k < N/2 \\ k - N & N/2 \leq k < N \end{cases} \quad (8)$$

not the raw index $0, 1, \dots, N-1$. Using the raw index gives a circular shift, which is exact only for integer $\hat{\tau}_s$; for fractional delays it adds a Hilbert-transform-like artifact to the negative-frequency content. The signed-index ramp is the band-limited (sinc-interpolated) delay, which is what we want.

One FFT, two uses. The same block FFT $R(k)$ feeds both the estimator (eq. (4)) and the corrector (eq. (7)). The correction adds only one complex multiply per bin plus an IFFT, so apart from a small per-block sin/cos lookup it is essentially free given the FFT that has already been computed for CD equalisation.

1.3 Block Boundaries and Overlap-Save

The phase-ramp correction in eq. (7) is a *circular* shift, not a linear one. The DFT treats the block as one period of an infinitely repeating sequence, so shifting by $\hat{\tau}_s$ samples means the IFFT output is

$$\text{outBlk}[n] = \text{block}[(n - \hat{\tau}_s) \bmod N]. \quad (9)$$

For a positive integer shift of $\hat{\tau}_s$ samples, the first $\hat{\tau}_s$ output samples come from the end of the input block (they wrapped around); for fractional shifts the same thing happens, just smeared across a few samples. These wrap-around samples are not the correctly-delayed signal — they are garbage stitched in from elsewhere. With consecutive blocks pasted edge-to-edge the result is a band of corrupted samples at every block boundary.

For a static offset, $\hat{\tau}_s$ is small (< 1 input sample for our test) so the corrupted band is a single sample and barely affects BER. With SFO, the cumulative unwrapped $\hat{\tau}_s$ grows linearly with the record: for SFO $\Delta f/f$ over L input samples,

$$\max |\hat{\tau}_s| \approx L \cdot \frac{\Delta f}{f} \quad \text{input samples}, \quad (10)$$

which can be tens or hundreds of samples for long records. Without protection, every block boundary corrupts $\sim |\hat{\tau}_s|$ samples and the BER floor is dominated by these wrap artifacts.

Overlap-save. The fix is to discard the corrupted boundary samples from every block and replace them with the clean middle of an adjacent block. The construction is:

1. Pad the input with G zeros at each end so that output sample 1 corresponds to input sample 1 and the final block does not run past the record.
2. Read blocks of N samples, stepping the read pointer by $M = N - 2G$ samples each block. Adjacent input blocks overlap by $2G$ samples.
3. In every IFFT output of length N , discard the first G and last G samples; keep the middle M . Write those M samples to the output, also stepping by M .

Because each IFFT-output discard region is replaced by the middle of the neighbouring block (which is shifted from a different chunk of input where its own discard region sits), the assembled output has no wrap-around contamination anywhere.

Choosing G . The guard must be larger than the worst-case wrap. After the running unwrap, $\hat{\tau}_s$ grows monotonically with the SFO drift across the record, so the binding constraint is

$$G \geq \max_b |\hat{\tau}_s(b)| \approx L \cdot \frac{\Delta f}{f}. \quad (11)$$

There is a throughput cost: each block of N samples produces only $M = N - 2G$ useful outputs, so the FFT work per output sample is inflated by $N/M = N/(N - 2G)$. Picking $G = N/4$ (the default) wastes 50% of the FFT work but tolerates a cumulative drift of

up to a quarter of the block size; for 100 ppm SFO over a record of 2^{18} symbols ($\eta L = 2^{19}$ samples) the maximum drift is ≈ 52 samples, comfortably under $G = 64$ for $N = 256$. For longer records or larger SFO either grow N (so $G = N/4$ is still enough) or pass a larger explicit G ; for very large drifts the more efficient option is to split $\hat{\tau}_s$ into an integer part absorbed into the input read pointer and a fractional part applied via the phase ramp, leaving only a G of ~ 1 sample.